

# Emergency Bypass Procedure: Pump P-101

Document ID: EBP-P101-2026

Effective Date: August 20, 2026

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## 1. Purpose

This procedure establishes the emergency bypass protocol for Centrifugal Pump P-101 during unplanned shutdown recovery scenarios.

## 2. Scope

Applies to all operations personnel responsible for restarting Pump P-101 after an emergency trip event.

## 3. Emergency Restart Procedure

### Step 2: Pressure Override Settings

- Set discharge pressure bypass to 145 PSI for initial surge clearance
- Maintain bypass at 145 PSI for minimum 3 minutes during startup transient
- Once flow stabilizes, reduce to normal operating pressure of 130 PSI

### Step 3: Vibration Monitoring

- Acceptable vibration during emergency restart: up to 8.0 mm/s RMS
- If vibration exceeds 8.0 mm/s, maintain current pressure and notify shift supervisor

### Step 4: Temperature Limits

- Maximum bearing temperature during emergency bypass: 95 degrees Celsius
- Normal operating temperature must not exceed 75 degrees Celsius

## 4. Safety Notes

- This procedure overrides normal operating limits ONLY during emergency restart
- Maximum duration of emergency bypass mode: 15 minutes
- All pressure settings validated for P-101 casing integrity

## 5. Reference Standards

- Safety Regulation SR-12: Valve Operations (max 100 PSI for bypass valves)
- P-101 Equipment Specifications
- OSHA 29 CFR 1910.119 Process Safety Management